

Practical Workbook
CT-265
Introduction to Tools and Techniques for
Data Science



Name: Moazzam Farooqui

Roll No: CT-24068

Batch: 2024

Department: CSIT

Department of Computer Science & Information Technology
NED University of Engineering & Technology

Table of Contents

S. No.	Object	Date
01	To build a strong foundation in Python programming by understanding the basic syntax, data types, operators, control flow, functions, and commonly used data structures.	9/4/2026
02	Learn how to use Pandas to make and work with Data Frames.	
03	To understand the NumPy library, which provides support for large, multi-dimensional arrays and matrices.	
04	To create effective static visualizations for data exploration and insight extraction.	
05	To apply foundational probability and statistics concepts to understand datasets.	
06	To provide a hands-on introduction to applying basic machine learning models for prediction.	
07	To acquire data from the web using web scraping techniques and extract structured information from websites.	
08	To apply natural language processing techniques to analyze and extract insights from customer review data.	
09	To understand the fundamentals of Microsoft Power BI for importing data, transforming it, building data models, creating visualizations, and designing interactive dashboards.	
10	To learn the basics of Tableau for connecting data, creating worksheets, applying filters, building dashboards, and presenting analytical stories from data.	

LAB # 01

CONTROL FLOW AND FUNCTIONS

NOTE: The screenshots and code snippets included in this document represent only the tasks that are required to be completed independently. They do not represent the complete Jupyter Notebook or all of its contents.

QUESTION NO # 01

if-elif-else Exercise

Write code to check weather condition with following scenarios:

- 1.) Sunny and Hot (temp is higher than 30)
- 2.) Sunny and Moderate (temp is between 20 and 30)
- 3.) Rainy and Cold (temp is less than 20)
- 4.) Rainy and Very Cold (temp is less than 5)

SOURCE CODE & OUTPUT:

```
# Write your code here
```

```
temp=float(input("Enter temperature in celsius:"))
```

```
if temp>30:
```

```
    print("It is sunny and hot")
```

```
elif temp>=20:
```

```
    print("It is sunny and moderate")
```

```
elif temp>=5:
```

```
    print("It is rainy and cold")
```

```
else:
```

```
    print("It is rainy and very cold")
```

```
It is sunny and moderate
```

QUESTION NO # 02

For loop Exercise

SOURCE CODE & OUTPUT:

```
#create empty list

result="result"
mylist=[]

#write for loop to assign each letter of "result" separately to list

for letter in result:
    mylist.append(letter)
```

```
#print list

print(mylist)

['r', 'e', 's', 'u', 'l', 't']
```

QUESTION NO # 03

While Loop Exercise

Write code to check if input number is greater than, less than or equal to 50. Loop should be continuous and command of **exit** should end the loop

Hint: you need to use input() function and cast the value to 'int' for conditions but don't cast when 'exit'.

The code for taking input and casting is as following:

```
x=input("Enter a your value: ")
```

SOURCE CODE & OUTPUT:

```
while True:
    m = input("Enter your value: ")

    if m.lower() == "exit":
        print("Exiting loop")
        break

    num = int(m)

    if num > 50:
        print("Number is greater than 50")
    elif num < 50:
        print("Number is less than 50")
    else:
        print("Number is equal to 50")
```

```
Number is greater than 50
Number is less than 50
Number is equal to 50
Exiting loop
```

QUESTION NO # 04

Home Task

Make a string having two words, starting with same letters and separated by a space like 'Abrar Ahmed' and perform the following tasks using methods

SOURCE CODE & OUTPUT:

```
#declare a string  
  
name=("Abrar Ahmed")
```

```
# split the string  
  
split_the_name=name.split(" ")  
print(split_the_name)
```

```
['Abrar', 'Ahmed']
```

```
# apply upper() method on the string  
  
upper_name=name.upper()  
print(upper_name)
```

```
ABRAR AHMED
```

```
# apply lower() method on the string  
  
lower_name=name.lower()  
print(lower_name)
```

```
abrar ahmed
```

QUESTION NO # 05

Functions exercise

Write a function to check if a number is even or odd.

SOURCE CODE & OUTPUT:

```
#define the function
def check_odd_or_even(num):
    if(num%2==0):
        print("This is an even number")
    else:
        print("This is an odd number")
```

```
#call the function
check_odd_or_even(19)
```

This is an odd number

QUESTION NO # 06

Task

Repeat the **filter()** function example with the **lambda()** functions

SOURCE CODE & OUTPUT:

```
# write the code here
```

```
list(filter(lambda num:num%2==0,nums))
```

```
[0, 2, 4, 6, 8, 10]
```


PYTHON OBJECTS AND DATA TYPES

QUESTION NO # 01

Task 1

Calculate the area of a rectangle having length of 15 cm and width of 10 cm. Use variable assignment and perform calculations using variables. Also display the result.

SOURCE CODE & OUTPUT:

```
# Assign values to variables
```

```
length=15  
width=10
```

```
# Perform Calculation
```

```
area_of_rectangle=length*width
```

```
# Display the result
```

```
print("AREA OF RECTANGLE IS:",area_of_rectangle)
```

```
AREA OF RECTANGLE IS: 150
```

QUESTION NO # 02

Task 2

Make a string having your name and print it 5 times.

SOURCE CODE & OUTPUT:

```
# string assignment  
  
name="Moazzam Farooqui"
```

```
# printing the string
```

```
for m in range(5):  
    print(name)
```

```
Moazzam Farooqui  
Moazzam Farooqui  
Moazzam Farooqui  
Moazzam Farooqui  
Moazzam Farooqui
```

QUESTION NO # 03

Indexing and Slicing

We know Lists are a sequence, which means Python can use indexes to call parts of the sequence. Let's learn how this works.

In Python, we use brackets `[]` after an object to call its index. We should also note that indexing starts at 0 for Python.

We can use a `:` to perform *slicing* which grabs everything up to a designated point.

Let's create a new object called `my_list` and then walk through a few examples of indexing.. Let's make a new list to remind ourselves of how this works:

SOURCE CODE & OUTPUT:

```
my_list = ['one','two','three',4,5]
```

Try yourself!

```
# Grab the second last index in reverse
```

```
my_list[-2]
```

QUESTION NO # 04

Task 3

Suppose we have a list containing areas of different rooms. Complete the given tasks using indexing and slicing.

SOURCE CODE & OUTPUT:

```
# The list having areas of 6 rooms respectively  
area=[28.3, 45.9, 123.4, 555, 213, 121]
```

```
# Show the area of third room in the list  
  
area[2]
```

```
123.4
```

```
# Show the areas of rooms first three rooms in the list  
  
area[:3]
```

```
[28.3, 45.9, 123.4]
```

```
# Show the area of rooms from 2 to 5  
area[1:5]
```

```
[45.9, 123.4, 555, 213]
```

QUESTION NO # 05

Task: Check type of 'key2'

SOURCE CODE & OUTPUT:

```
my_dict = {'key1':123,'key2':[12,23,33],'key3':['item0','item1','item2']}
```

```
# Try here!  
type(my_dict['key2'])
```

```
list
```

QUESTION NO # 06

Dictionaries Exercise

SOURCE CODE & OUTPUT:

```
# Definition of countries and capital
countries = ['spain', 'france', 'germany', 'norway']
capitals = ['madrid', 'paris', 'berlin', 'oslo']
```

```
# From string in countries and capitals, create dictionary europe

europe = {
    countries[0]: capitals[0],
    countries[1]: capitals[1],
    countries[2]: capitals[2],
    countries[3]: capitals[3]
}

#print europe

print(europe)

{'spain': 'madrid', 'france': 'paris', 'germany': 'berlin', 'norway': 'oslo'}
```

QUESTION NO # 07

Object Type Casting

You can cast type of any object in Python. Common data types casting functions include:

- **int()** (for integer)
- **float()**
- **str()** (for string)
- **bool()** (for Boolean True/False)

You can type cast any object through the following code. The given example is converting **float** to **int**.

```
int(5.8)
```

Home Task

SOURCE CODE & OUTPUT:

```
# convert 80 into float type
```

```
float(80)
```

```
80.0
```

```
# convert 20.9 into a string and check its type
```

```
num=str(20.9)  
print(type(num))
```

```
<class 'str'>
```

```
# convert a boolean value into int
```

```
int(True)
```

```
1
```

```
# convert '123' into float
```

```
float('123')
```

```
123.0
```